

DIABETIC RETINOPATHY SCREENING REPORT

AI-Assisted Fundus Image Analysis — with Bootstrap 95% CI

Image: idrid_IDRiD_1196.jpg
Date: 2026-03-28 12:20:28
Model: EfficientNet-B3 (Binary + Multiclass Cascade)
Bootstrap: 2000 resamples, 95% confidence level

SCREENING RESULT (with 95% Bootstrap CI)

Binary Detection:

- Ensemble (5-fold+TTA): DR Detected
P(DR) = 0.9762 95% CI [0.9687, 0.9826]
- 2-Phase (single+TTA): DR Detected
P(DR) = 0.9985 95% CI [0.9976, 0.9993]

Severity Grading (Direct Multiclass Model):

- Predicted Grade: 4 — Proliferative DR
- Confidence: 64.4% 95% CI [0.5577, 0.7215]
- Severity: Sight-threatening

Per-Grade Probabilities (95% CI):

G0 (No DR)	): 0.0232	[0.0150, 0.0325]
G1 (Mild NPDR)	): 0.0293	[0.0176, 0.0444]
G2 (Moderate NPDR)	): 0.1034	[0.0784, 0.1320]
G3 (Severe NPDR)	): 0.2005	[0.1539, 0.2506]
G4 (Proliferative DR)	): 0.6436	[0.5577, 0.7215] ◀

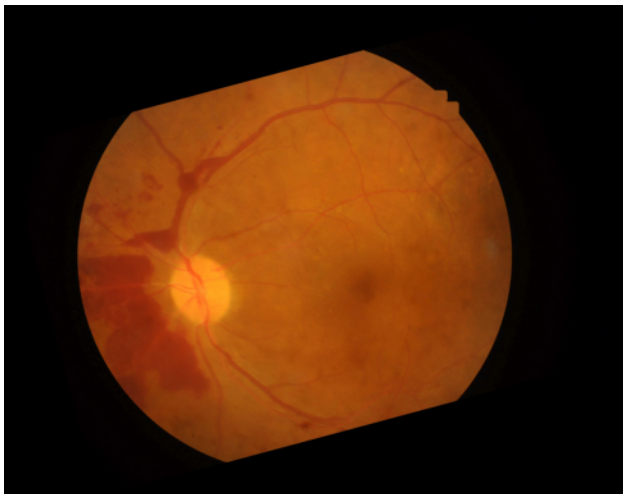
Clinical Recommendation:

- **URGENT referral — risk of vision loss**

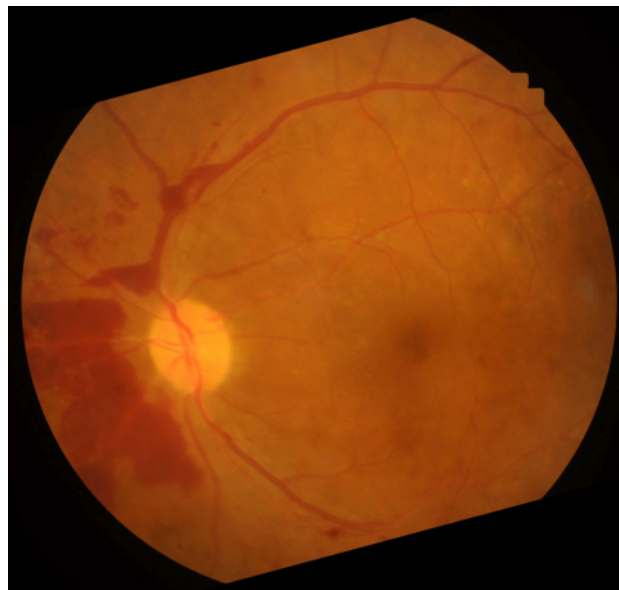
DISCLAIMER: This is an AI screening aid, not a diagnostic tool. All results must be confirmed by a qualified ophthalmologist.

Preprocessing Pipeline — idrid_IDRiD_1196.jpg

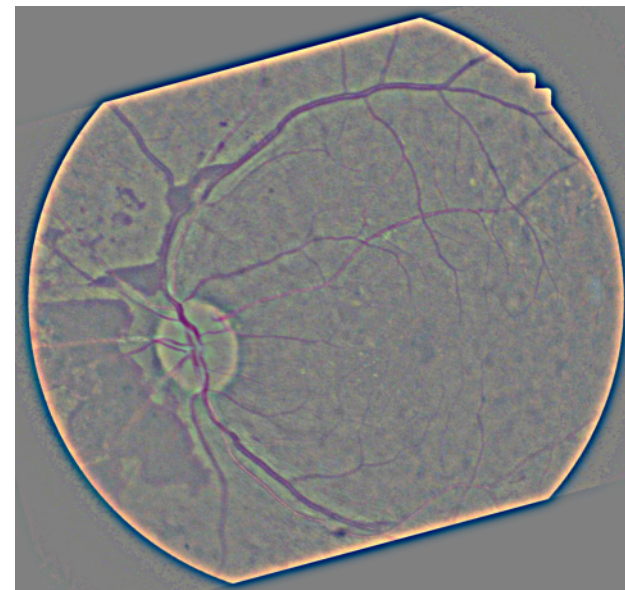
Step 1: Original



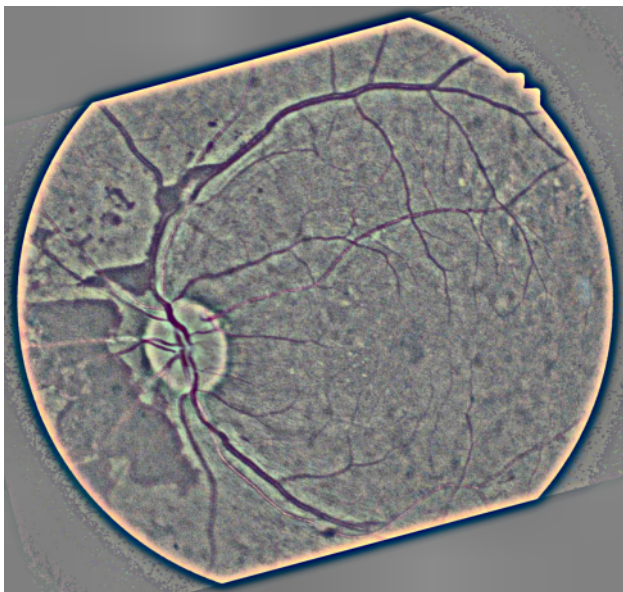
Step 2: Circle Crop



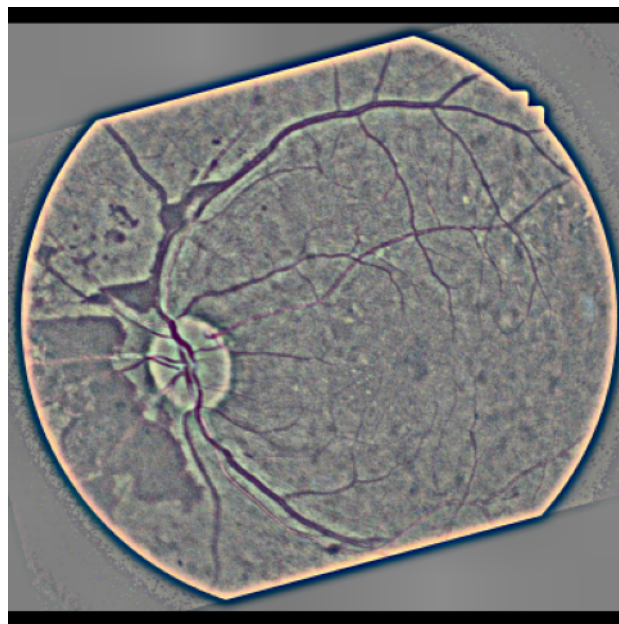
Step 3: Ben Graham



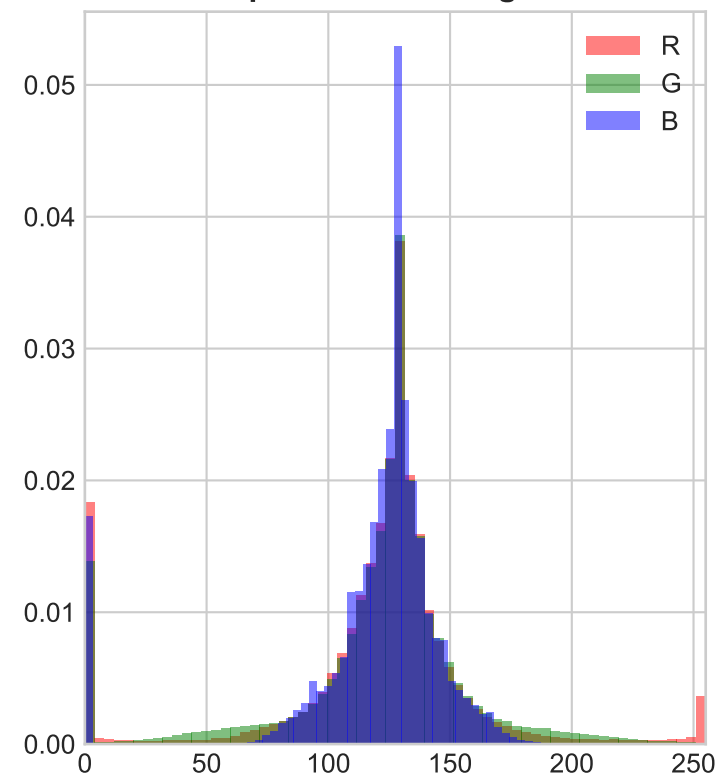
Step 4: CLAHE



Step 5: Final (384×384)

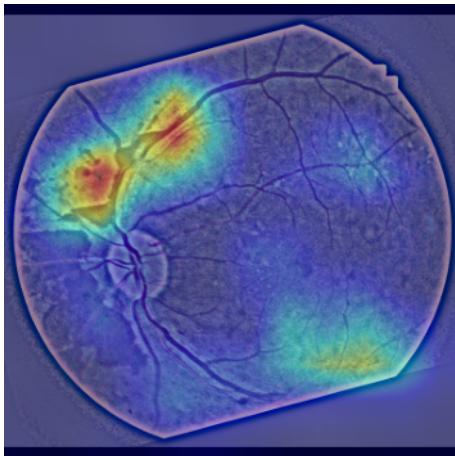


Preprocessed Histograms

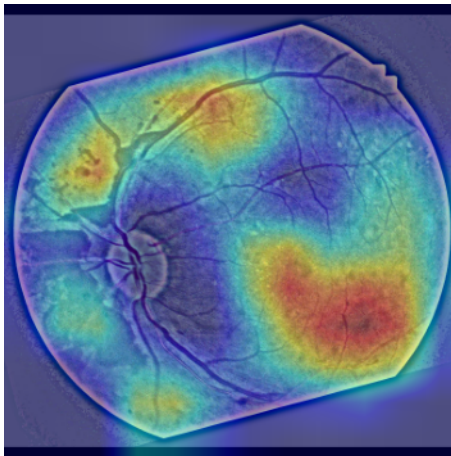


Explainability & Predictions with 95% CI — idrid_IDRiD_1196.jpg

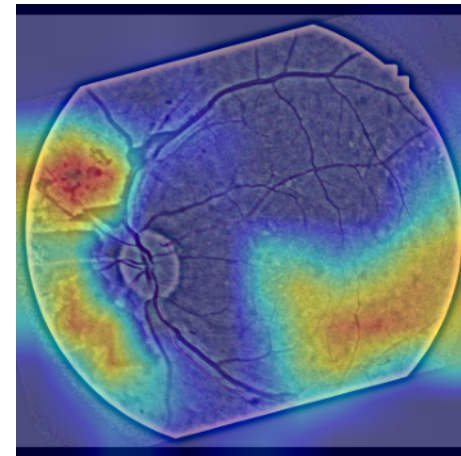
Ensemble GradCAM
P(DR)=0.9762 CI[0.9687,0.9826]



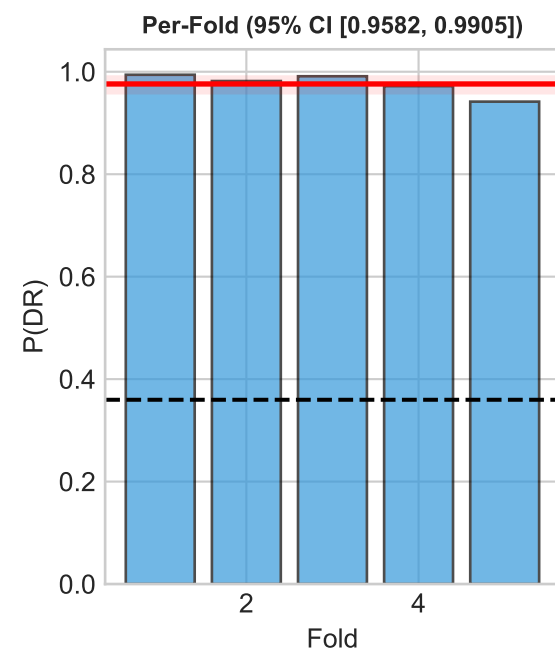
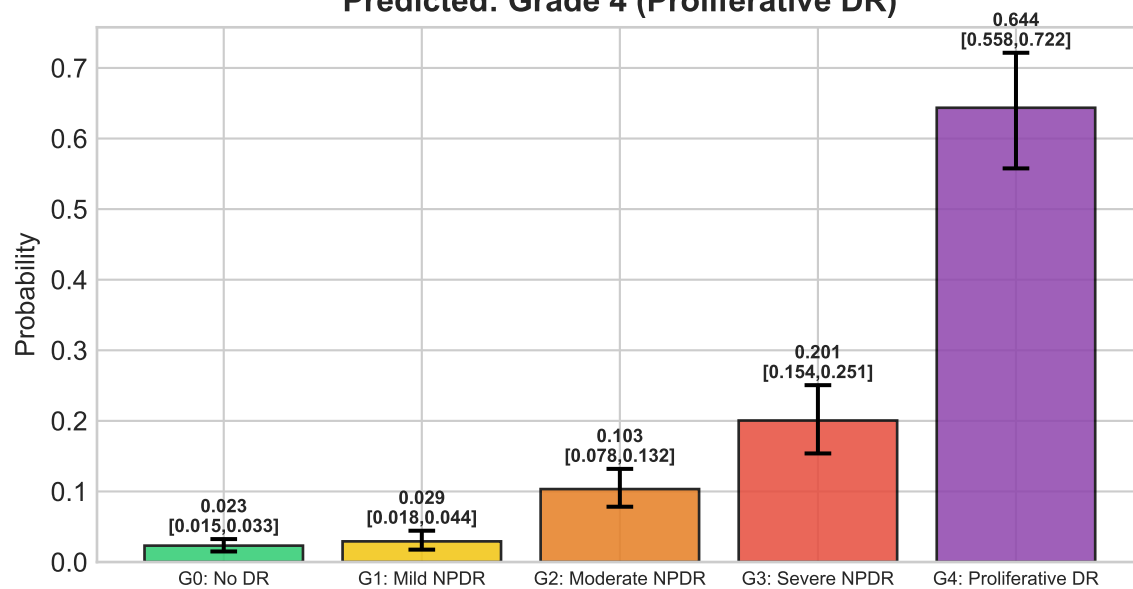
2-Phase GradCAM
P(DR)=0.9985 CI[0.9976,0.9993]



Multiclass GradCAM (G4)
Conf=64.36% CI[0.558,0.722]



5-Grade Probabilities with 95% CI (n_boot=2000)
Predicted: Grade 4 (Proliferative DR)



Model Validation Summary

MODEL VALIDATION METRICS

Dataset: APTOS 2019 (531 test images)

Bootstrap: 2000 resamples, 95% CI

Metric	Ensemble+TTA	2-Phase+TTA
AUC-ROC	0.9992	0.9992
Accuracy	0.9906	0.9887
Sensitivity	0.9962	0.9962
Specificity	0.9852	0.9815
F1-Score	0.9905	0.9886
Cohen Kappa	0.9812	0.9774

Multiclass Model (Direct 5-Class):

Best Validation QWK: 0.859

Architecture: EfficientNet-B3 + Dropout(0.4) + Linear(1536, 5)

Training: CrossEntropyLoss + label_smoothing=0.1

THIS IMAGE – PREDICTION SUMMARY

Ensemble P(DR): 0.976153 95% CI [0.968707, 0.982596]

2-Phase P(DR): 0.998516 95% CI [0.997609, 0.999289]

Multiclass Grade: 4 (Proliferative DR)

Confidence: 0.6436 95% CI [0.5577, 0.7215]

Per-Grade Probabilities (95% CI):

G0 No DR : 0.0232 [0.0150, 0.0325]

G1 Mild NPDR : 0.0293 [0.0176, 0.0444]

G2 Moderate NPDR : 0.1034 [0.0784, 0.1320]

G3 Severe NPDR : 0.2005 [0.1539, 0.2506]

G4 Proliferative DR : 0.6436 [0.5577, 0.7215]